V- L'Hôpital rule

Note that: By using $\overset{"L"}{=}$, we mean that we apply L'Hôpital rule

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a) $\lim\limits_{x\to 0}\dfrac{\sin x}{x}=\dfrac{0}{0}$

$$\overset{L}{=}\lim_{x\to 0}\frac{\cos x}{1}=\cos(0)=1$$

(b) $\lim\limits_{x\to 0}\dfrac{a^x-b^x}{c^x-d^x}=\dfrac{a^0-b^0}{c^0-d^0}=\dfrac{1-1}{1-1}=\dfrac{0}{0}$

$$\overset{L}{=}\lim_{x\to 0}\frac{a^x\ln a-b^x\ln b}{c^x\ln c-d^x\ln d}=\frac{\ln a-\ln b}{\ln c-\ln d}$$

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c) $\lim\limits_{x\to 0}\dfrac{\ln(\cos 2x)}{\sin(2x)}=\dfrac{\ln(\cos 0)}{\sin 0}=\dfrac{\ln(1)}{0}=\dfrac{0}{0}$

$$\overset{L}{=}\lim_{x\to 0}\frac{\dfrac{-2\sin(2x)}{\cos(2x)}}{2\cos(2x)}=\lim_{x\to 0}\frac{-\sin(2x)}{\cos^2(2x)}=\frac{0}{1}=0$$

(d) $\lim\limits_{x\to 0^+}\dfrac{\ln x}{1+2\ln(\sin x)}=\dfrac{-\infty}{-\infty}$

$$\overset{L}{=}\lim_{x\to 0^+}\frac{\dfrac{1}{x}}{\dfrac{2\cos x}{\sin x}}=\lim_{x\to 0^+}\frac{\sin x}{2x\cos x}=\frac{1}{2}\lim_{x\to 0^+}\frac{\sin x}{x}\lim_{x\to 0^+}\frac{1}{\cos x}=\frac{1}{2}.(1).(1)=\frac{1}{2}$$

(e) $\lim\limits_{x\to 1}\dfrac{x^9-1}{x^2-1}=\dfrac{0}{0}$

$$\overset{L}{=}\lim_{x\to 1}\frac{9x^8}{2x}=\frac{9}{2}$$

(f) $\lim\limits_{x\to\frac{\pi}{2}^+}\dfrac{\cos x}{1-\sin x}=\dfrac{0}{0}$

$$\overset{L}{=}\lim_{x\to\frac{\pi}{2}^+}\frac{-\sin x}{-\cos x}=\lim_{x\to\frac{\pi}{2}^+}\tan x=-\infty$$

(g) $\lim_{x\to 0}\frac{a^x-1}{x}=\frac{0}{0}$ **this is a second solution for the problem 13 in ws4**

$$\overset{L}{=}\lim_{x\to 0}\frac{a^x\ln a}{1}=a^0\ln a=\ln a$$

(h) $\lim_{x\to\infty}\frac{e^x}{x}=\frac{\infty}{\infty}$

$$\overset{L}{=}\lim_{x\to\infty}\frac{e^x}{1}=\infty$$

(i) $\lim_{x\to\infty}\frac{\ln(\ln x)}{x}=\frac{\infty}{\infty}$

$$\overset{L}{=}\lim_{x\to\infty}\frac{\frac{1}{\ln x}.\frac{1}{x}}{1}=\lim_{x\to\infty}\frac{1}{x\ln x}=\frac{1}{\infty}=0$$

(j) $\lim_{x\to 0}\frac{e^x-1-x}{x^2}=\frac{0}{0}$

$$\overset{L}{=}\lim_{x\to 0}\frac{e^x-1}{2x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 0}\frac{e^x}{2}=\frac{1}{2}$$

(k) $\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{0}{0}$ **this is a second solution for the problem 8 in ws4**

$$\overset{L}{=}\lim_{x\to 0}\frac{\sin x}{2x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 0}\frac{\cos x}{2}=\frac{1}{2}$$

(l) $\lim_{x\to 0}\frac{x}{\tan^{-1}(4x)}=\frac{0}{0}$

$$\overset{L}{=}\lim_{x\to 0}\frac{1}{\frac{1}{1+(4x)^2}.4}=\frac{1}{4}$$

(m) $\lim_{x\to 0^+}\sqrt{x}.\ln(x)=0.(-\infty)$ **write the product as quotient**

$$=\lim_{x\to 0^+}\frac{\ln x}{\frac{1}{\sqrt{x}}}=\lim_{x\to 0^+}\frac{\ln x}{x^{-\frac{1}{2}}}\frac{-\infty}{\infty}$$

$$\overset{L}{=}\lim_{x\to 0^+}\frac{\frac{1}{x}}{-\frac{1}{2}x^{\frac{-3}{2}}}=\lim_{x\to 0^+}-2x^{\frac{1}{2}}=0$$

(n) $\lim_{x\to 0^+} \sin x.\ln x = 0.(-\infty)$ **write the product as quotient**

$$= \lim_{x\to 0^+} \frac{\ln x}{1/\sin x} = \lim_{x\to 0^+} \frac{\ln x}{\csc x} = \frac{-\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\csc x \cot x} = \lim_{x\to 0^+} \frac{\sin^2 x}{x\cos x} = \lim_{x\to 0^+} \frac{\sin x}{x} . \lim_{x\to 0^+} \frac{\sin x}{\cos x} = (1).(0) = 0$$

(o) $\lim_{x\to 0}\left[(e^x + e^{-x} - 2).\cot x\right] = 0.(\infty)$ **write the product as quotient**

$$= \lim_{x\to 0} \frac{(e^x + e^{-x} - 2)}{\tan x} = \frac{0}{0}$$

$$\overset{L}{=} \mathbf{\lim_{x\to 0} \frac{e^x - e^{-x}}{\sec^2 x} = \frac{0}{1} = 0}$$

(p) $\lim_{x\to 1^+}\left[\ln x.\ln(x-1)\right] = 0.(-\infty)$ **write the product as quotient**

$$= \lim_{x\to 1^+} \frac{\ln(x-1)}{\frac{1}{\ln(x)}} = \lim_{x\to 1^+} \frac{\ln(x-1)}{(\ln(x))^{-1}} = \frac{-\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to 1^+} \frac{\frac{1}{x-1}}{-(\ln x)^{-2}\left(\frac{1}{x}\right)} = \lim_{x\to 1^+} \frac{-x(\ln x)^2}{x-1} = \frac{0}{0}$$

$$\overset{L}{=} \lim_{x\to 1^+} \frac{-(\ln x)^2 - 2x\ln x\left(\frac{1}{x}\right)}{1} = \lim_{x\to 1^+} -(\ln x)^2 - 2\ln x = 0$$

(q) $\lim_{x\to\infty} x\tan\left(\frac{1}{x}\right) = \infty.0$ **write the product as quotient**

$$= \lim_{x\to\infty} \frac{\tan\left(\frac{1}{x}\right)}{\frac{1}{x}} = \frac{0}{0}$$

$$\overset{L}{=} \lim_{x\to\infty} \frac{\sec^2\left(\frac{1}{x}\right).\left(\frac{-1}{x^2}\right)}{\left(\frac{-1}{x^2}\right)} = \lim_{x\to\infty} \sec^2\left(\frac{1}{x}\right) = 1$$

(r) $\lim\limits_{x\to 1}\left[\sin(x-1).\tan\dfrac{\pi x}{2}\right]=0.(\pm\infty)$ **write the product as quotient**

$$\lim_{x\to 1}\left[\sin(x-1).\tan\frac{\pi x}{2}\right]=\lim_{x\to 1}\frac{\sin(x-1)}{\cot\left(\frac{\pi x}{2}\right)}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 1}\frac{\cos(x-1)}{-\csc^2\left(\frac{\pi x}{2}\right)\left(\frac{\pi}{2}\right)}=\lim_{x\to 1}\left(-\frac{2}{\pi}\right)\left(\sin^2\left(\frac{\pi x}{2}\right)\right)(\cos(x-1))=-\frac{2}{\pi}$$

or

$$\lim_{x\to 1}\left[\sin(x-1).\tan\frac{\pi x}{2}\right]=\lim_{x\to 1}\frac{\sin(x-1)\sin\left(\frac{\pi x}{2}\right)}{\cos\left(\frac{\pi x}{2}\right)}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 1}\frac{\cos(x-1)\sin\left(\frac{\pi x}{2}\right)+\frac{\pi}{2}\sin(x-1)\cos\left(\frac{\pi x}{2}\right)}{-\frac{\pi}{2}\sin\left(\frac{\pi x}{2}\right)}=\frac{\cos(0)\sin\left(\frac{\pi}{2}\right)+\frac{\pi}{2}\sin(0)\cos\left(\frac{\pi}{2}\right)}{-\frac{\pi}{2}\sin\left(\frac{\pi}{2}\right)}=\frac{1}{-\frac{\pi}{2}}=-\frac{2}{\pi}$$

(s) $\lim\limits_{x\to 0}\left(\dfrac{1}{x}-\csc x\right)=\infty-\infty$ **write the difference as quotient**

$$=\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{\sin x}\right)=\lim_{x\to 0}\frac{\sin x-x}{x\sin x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 0}\frac{\cos x-1}{x\cos x+\sin x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 0}\frac{-\sin x}{-x\sin x+\cos x+\cos x}=0$$

(t) $\lim\limits_{x\to 1}\left(\dfrac{1}{\ln x}-\dfrac{1}{x-1}\right)=\infty-\infty$ **write the difference as quotient**

$$=\lim_{x\to 1}\frac{(x-1)-\ln x}{(x-1)\ln x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 1}\frac{1-\frac{1}{x}}{(x-1)\frac{1}{x}+\ln x}\overset{(*x)}{=}\lim_{x\to 1}\frac{x-1}{(x-1)+x\ln x}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to 1}\frac{1}{1+x\left(\frac{1}{x}\right)+\ln x}=\frac{1}{2}$$

(u) $\lim_{x\to\infty}\sqrt{x^2+x}-x=\infty-\infty$ **(this solution is without L'Hopital rule)**

$$\lim_{x\to\infty}\sqrt{x^2+x}-x=\lim_{x\to\infty}\frac{(\sqrt{x^2+x}-x)(\sqrt{x^2+x}+x)}{(\sqrt{x^2+x}+x)}=\lim_{x\to\infty}\frac{x^2+x-x^2}{\sqrt{x^2+x}+x}=\lim_{x\to\infty}\frac{x}{\sqrt{x^2+x}+x}=\frac{\infty}{\infty}$$

$$\lim_{x\to\infty}\frac{x}{\sqrt{x^2+x}+x}\overset{(\div x)}{=}\lim_{x\to\infty}\frac{1}{\sqrt{1+\frac{1}{x}}+1}=\frac{1}{2}$$

Or **(by using L'Hopital rule)** **write the difference as quotient**

$$\lim_{x\to\infty}\sqrt{x^2+x}-x=\lim_{x\to\infty}\sqrt{x^2+x}-\sqrt{x^2}=\lim_{x\to\infty}\sqrt{x^2}\left[\sqrt{1+\frac{1}{x}}-1\right]=\lim_{x\to\infty}\frac{\sqrt{1+\frac{1}{x}}-1}{\frac{1}{x}}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to\infty}\frac{\frac{1}{2\sqrt{1+\frac{1}{x}}}\cdot\left(\frac{-1}{x^2}\right)}{\left(\frac{-1}{x^2}\right)}=\frac{1}{2}$$

(v) $\lim_{x\to\infty}\left(xe^{\frac{1}{x}}-x\right)=\infty-\infty$ **take x as common factor**

$$=\lim_{x\to\infty}x\left(e^{\frac{1}{x}}-1\right)=\lim_{x\to\infty}\frac{e^{\frac{1}{x}}-1}{x^{-1}}=\frac{0}{0}$$

$$\overset{L}{=}\lim_{x\to\infty}\frac{e^{\frac{1}{x}}(-x^{-2})}{(-x^{-2})}=\lim_{x\to\infty}e^{\frac{1}{x}}=1$$

(w) $\lim_{x\to\infty}[x-\ln^3 x]=\infty-\infty$ **take x as common factor**

$$=\lim_{x\to\infty}x\left[1-\frac{\ln^3 x}{x}\right]=\lim_{x\to\infty}x.\lim_{x\to\infty}\left[1-\frac{\ln^3 x}{x}\right]=\lim_{x\to\infty}x.\left[\lim_{x\to\infty}1-\underbrace{\lim_{x\to\infty}\frac{\ln^3 x}{x}}_{?}\right]=\infty[1-?]\Rightarrow(1)$$

$$?=\lim_{x\to\infty}\frac{\ln^3 x}{x}=\frac{\infty}{\infty}\qquad\overset{L}{=}\lim_{x\to\infty}\frac{3\ln^2 x\left(\frac{1}{x}\right)}{1}=\lim_{x\to\infty}\frac{3\ln^2 x}{x}=\frac{\infty}{\infty}$$

$$\overset{L}{=}\lim_{x\to\infty}\frac{6\ln x\left(\frac{1}{x}\right)}{1}=\lim_{x\to\infty}\frac{6\ln x}{x}=\frac{\infty}{\infty}\qquad\overset{L}{=}\lim_{x\to\infty}\frac{6\left(\frac{1}{x}\right)}{1}=\lim_{x\to\infty}\frac{6}{x}=\frac{6}{\infty}=0$$

$$\overset{(1)}{\Rightarrow}\lim_{x\to\infty}x\left[1-\frac{\ln^3 x}{x}\right]=\infty.[1-0]=\infty(1)=\infty$$

Remark: if $\lim_{x\to a}[f(x)-g(x)\ln x]=\infty-\infty=\lim_{x\to a}f(x)\left[1-\frac{g(x)\ln x}{f(x)}\right]$

(x) $\lim\limits_{x\to 0^+} x^{x^2} = 0^0$

Let $y = x^{x^2} \qquad \Rightarrow \ln y = x^2 \ln x$

$$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} x^2 \ln x = 0.(-\infty)$$

$$= \lim_{x\to 0^+} \frac{\ln x}{x^{-2}} = \frac{-\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to 0^+} \frac{\frac{1}{x}}{-2x^{-3}} = \lim_{x\to 0^+} \frac{-x^2}{2} = 0$$

$$\ln\left(\lim_{x\to 0^+} y\right) = 0 \Rightarrow \lim_{x\to 0^+} y = e^0$$

$$\therefore \lim_{x\to 0^+} x^{x^2} = e^0 = 1$$

(y) $\lim\limits_{x\to 0^+} (\tan 2x)^x = 0^0$

Let $y = (\tan 2x)^x \quad \Rightarrow \ln y = x \ln(\tan 2x)$

$$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} x.\ln(\tan 2x) = 0.(-\infty)$$

$$\boldsymbol{= \lim_{x\to 0^+} \frac{\ln[\tan(2x)]}{x^{-1}} = \frac{-\infty}{\infty}}$$

$$\overset{L}{=} \lim_{x\to 0^+} \frac{\frac{\sec^2(2x).(2)}{\tan(2x)}}{-x^{-2}} = \lim_{x\to 0^+} \frac{x}{\tan(2x)}.(-2x\sec^2(2x))$$

$$= \left[\lim_{x\to 0^+} \frac{1}{2}\frac{2x}{\tan(2x)}\right].\left[\lim_{x\to 0^+} -2x\sec^2(2x)\right] = \left[\frac{1}{2}\right].[0] = 0$$

$$\therefore \lim_{x\to 0^+} (\tan(2x))^x = e^0 = 1$$

(z) $\lim\limits_{x\to 0} (1-2x)^{\frac{1}{x}} = 1^\infty$

Let $y = (1-2x)^{\frac{1}{x}} \quad \Rightarrow \ln y = \frac{1}{x}.\ln(1-2x)$

$$\lim_{x\to 0} \ln y = \lim_{x\to 0} \frac{\ln(1-2x)}{x} = \frac{0}{0}$$

$$\overset{L}{=} \lim_{x\to 0} \frac{\ln(1-2x)}{x} = \lim_{x\to 0} \frac{\frac{-2}{1-2x}}{1} = -2$$

$$\therefore \lim_{x\to 0} (1-2x)^{\frac{1}{x}} = e^{-2}$$

(i) $\lim_{x\to\infty} x^{\frac{\ln 2}{1+\ln x}} = \infty^0$

Let $y = x^{\frac{\ln 2}{1+\ln x}} \quad \Rightarrow \ln y = \frac{\ln 2}{1+\ln x}.\ln x$

$$\lim_{x\to\infty} \ln y = \lim_{x\to\infty} \frac{\ln 2.\ln x}{1+\ln x} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \ln 2 \lim_{x\to\infty} \frac{\frac{1}{x}}{\frac{1}{x}} = \ln 2$$

$$\therefore \lim_{x\to\infty} x^{\frac{\ln 2}{1+\ln x}} = e^{\ln 2} = 2$$

(ii) $\lim_{x\to\infty} (e^x + x)^{\frac{1}{x}} = \infty^0$

Let $y = (e^x + x)^{\frac{1}{x}} \quad \Rightarrow \ln y = \frac{1}{x}.\ln(e^x + x)$

$$\lim_{x\to\infty} \ln y = \lim_{x\to\infty} \frac{\ln(e^x + x)}{x} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to\infty} \frac{\frac{e^x + 1}{e^x + x}}{1} = \lim_{x\to\infty} \frac{e^x + 1}{e^x + x} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to\infty} \frac{e^x}{e^x + 1} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to\infty} \frac{e^x}{e^x} = \lim_{x\to\infty} 1 = 1$$

$$\therefore \lim_{x\to\infty} (e^x + x)^{\frac{1}{x}} = e^1 = e$$

(iii) $\lim_{x\to 0} (\cos 3x)^{\frac{5}{x}} = 1^\infty$

Let $y = (\cos 3x)^{\frac{5}{x}} \quad \Rightarrow \ln y = \frac{5}{x}.\ln(\cos 3x)$

$$\lim_{x\to 0} \ln y = \lim_{x\to 0} \frac{5\ln(\cos 3x)}{x} = \frac{0}{0}$$

$$\overset{L}{=} \lim_{x\to 0} \frac{\frac{-15\sin(3x)}{\cos(3x)}}{1} = \lim_{x\to 0} (-15\tan(3x)) = 0$$

$$\therefore \lim_{x\to 0} (\cos 3x)^{\frac{5}{x}} = e^0 = 1$$

(iv) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = 1^\infty$

Let $y=\left(1+\frac{1}{x}\right)^x \quad \Rightarrow \ln y = x.\ln\left(1+\frac{1}{x}\right)$

$$\lim_{x\to\infty}\ln y = \lim_{x\to\infty} x\ln\left(1+\frac{1}{x}\right) = \lim_{x\to\infty}\frac{\ln\left(1+\frac{1}{x}\right)}{\frac{1}{x}} = \frac{0}{0}$$

$$\overset{L}{=} \lim_{x\to\infty}\frac{\frac{1}{\left(1+\frac{1}{x}\right)}\left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)} = \lim_{x\to\infty}\frac{1}{\left(1+\frac{1}{x}\right)} = \frac{1}{1+0} = 1$$

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e^1$$

(*) $\lim_{x\to\infty}\left[\sqrt{x}\ln x - x^2.\ln(\ln x)\right] = \infty-\infty$ **(using the remark we mentioned in (w) in this problem)**

(take $\sqrt{x}\ln x$ as common factor)

$$= \lim_{x\to\infty}\sqrt{x}\ln x\left[1-\frac{x^{\frac{3}{2}}\ln(\ln x)}{\ln x}\right] = \lim_{x\to\infty}\sqrt{x}\ln x\left[1-\underbrace{\lim_{x\to\infty}\frac{x^{\frac{3}{2}}\ln(\ln x)}{\ln x}}_{?}\right] = \infty[1-?] \Rightarrow (1)$$

$$? = \lim_{x\to\infty}\frac{x^{\frac{3}{2}}\ln(\ln x)}{\ln x} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to\infty}\frac{\frac{3}{2}x^{\frac{1}{2}}\ln(\ln x) + x^{\frac{3}{2}}\frac{1}{\ln x}\frac{1}{x}}{\frac{1}{x}} = \lim_{x\to\infty}\frac{3}{2}x^{\frac{3}{2}}\ln(\ln x) + \frac{x^{\frac{3}{2}}}{\ln x} = \infty + \underbrace{\lim_{x\to\infty}\frac{x^{\frac{3}{2}}}{\ln x}}_{??} = \infty + ?? \qquad \Rightarrow (2)$$

$$?? = \lim_{x\to\infty}\frac{x^{\frac{3}{2}}}{\ln x} = \frac{\infty}{\infty}$$

$$\overset{L}{=} \lim_{x\to\infty}\frac{x^{\frac{3}{2}}}{\ln x} = \lim_{x\to\infty}\frac{\frac{3}{2}x^{\frac{1}{2}}}{\frac{1}{x}} = \lim_{x\to\infty}\frac{3}{2}x^{\frac{3}{2}} = \infty$$

$$\overset{(2)}{\Rightarrow} \quad ? = \infty+\infty = \infty$$

$$\overset{(1)}{\Rightarrow} \quad \lim_{x\to\infty}\left[\sqrt{x}\ln x - x^2.\ln(\ln x)\right] = \infty[1-\infty] = -\infty$$

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